Introduction to Markdown & LaTeX

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Outline

Markdown

LaTeX

Markdown

What is Markdown?

- Markdown is a lightweight markup language.
- It uses plain text formatting syntax to convert text to HTML (Web pagaes).
- Ideal for creating documentation, reports, and readme files.

Markdown Example

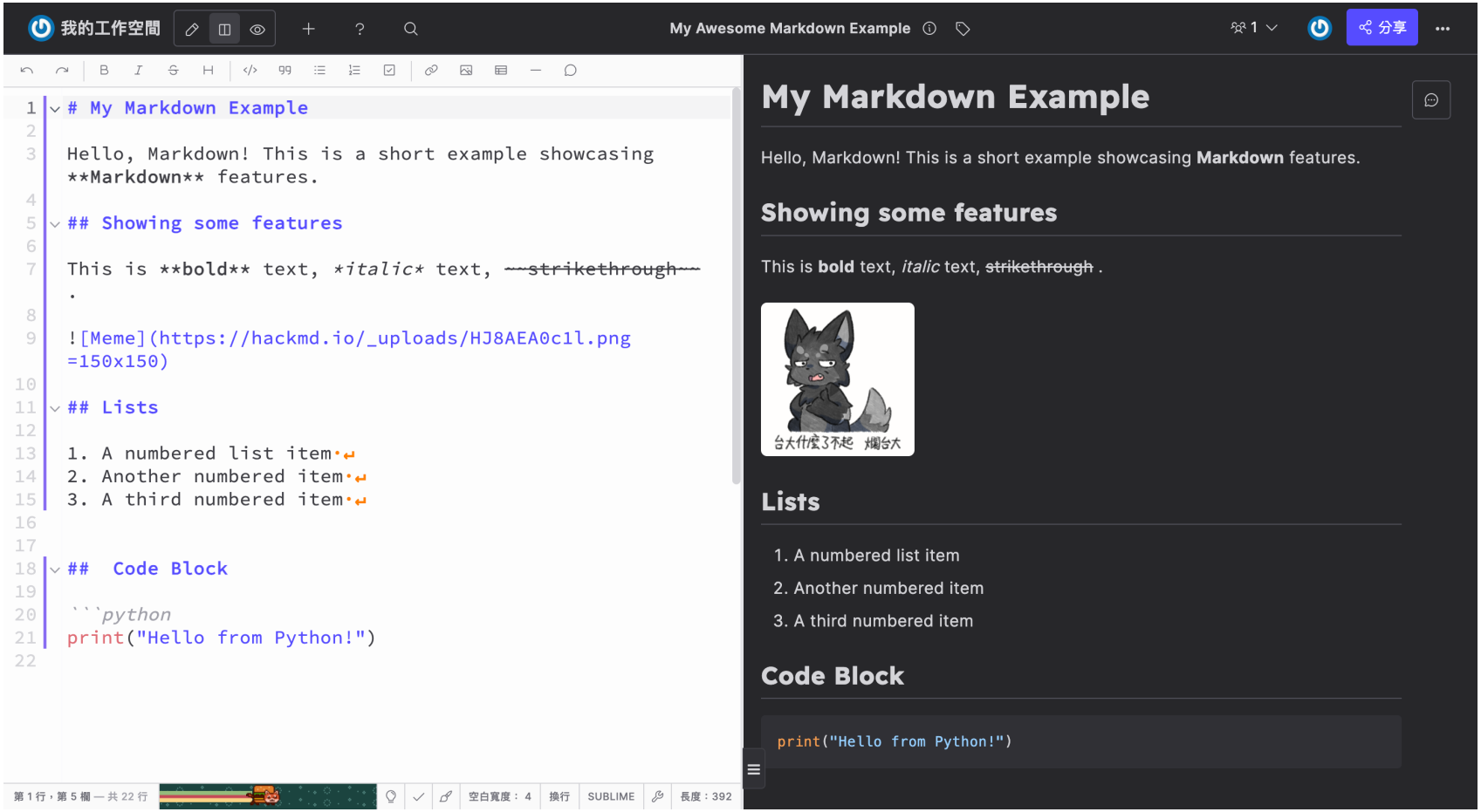


Figure: Example of Markdown formatting and its rendered output

Basic Markdown Syntax

Headers

```
# Header 1

## Header 2

### Header 3
```

Emphasis

`*italic*` or `_italic_`

`**bold**` or `__bold__`

Tables and Math in Markdown

Tables

```
| Column 1 | Column 2 |
|----------|----------|
| Cell 1   | Cell 2   |
| Cell 3   | Cell 4   |
```

Math Expressions

- Inline math: `$E = mc^2$`
- Display math: `$$\sum_{i=1}^n x_i = \bar{x}$$`
- Supports most LaTeX math syntax

Code Blocks in Markdown

- Inline code: ‘`code`‘
- Multi-line code blocks: Use triple backticks (‘‘‘) before and after the code.

Usage of Markdown

- Documentation: Clear written explanations of code, methodologies, and project workflows.
- Note-taking: Efficient alternative to MS Word for class notes with simple formatting.
- Supports structured content with headings, lists, and code examples.
- Popular in data science platforms like GitHub, Jupyter Notebooks, and R Markdown for sharing analyses.
- Static website generation: Markdown can be used with static site generators like Hexo to create personal blogs and websites (e.g., `https://aaronlin1229.github.io/`).

Where to Write Markdown & What to Write

HackMD (`https://hackmd.io`)

- Real-time collaborative markdown editor
- Perfect for:
  - Taking class notes
  - Writing project documentation
  - Creating shared meeting notes

Common Document Types

- README files for code repositories
- Technical documentation
- Project wikis
- Course notes
- Blog posts
- Quick documentation that needs web rendering

LaTeX

What is LaTeX?

- LaTeX is the standard typesetting system for writing academic papers and research articles.
- It excels at formatting mathematical formulas, tables, figures, and citations needed in papers.
- Most academic journals require or strongly prefer LaTeX submissions.
- You should notice a lot of your problem sets and tests from other courses are typesetted using LaTeX.

Benefits:

- Produces high-quality, publication-ready documents.
- Superior handling of equations, tables, and figures.
- Ensures consistency in formatting across long documents.

Basic LaTeX Structure

Example

```
\documentclass{article}
\begin{document}
Your content here...
\end{document}
```

Writing Equations in LaTeX

LaTeX Code

```
Inline equation: \( V^{\pi*}(s) \)
Displayed equation:
\[
V^{\pi*}(s)=  \max_a \left\{ {R(s,a) +
\gamma \sum_{s'} P(s'|s,a) V^{\pi*}(s')} \right\}
\]
```

Rendered Output

- Inline equation: $V^{\pi*}(s)$
- Displayed equation:

$$V^{\pi*}(s)= \max_a \left\{ R(s,a) + \gamma \sum_{s'} P(s'|s,a) V^{\pi*}(s') \right\}$$

Creating Tables in LaTeX

LaTeX Code

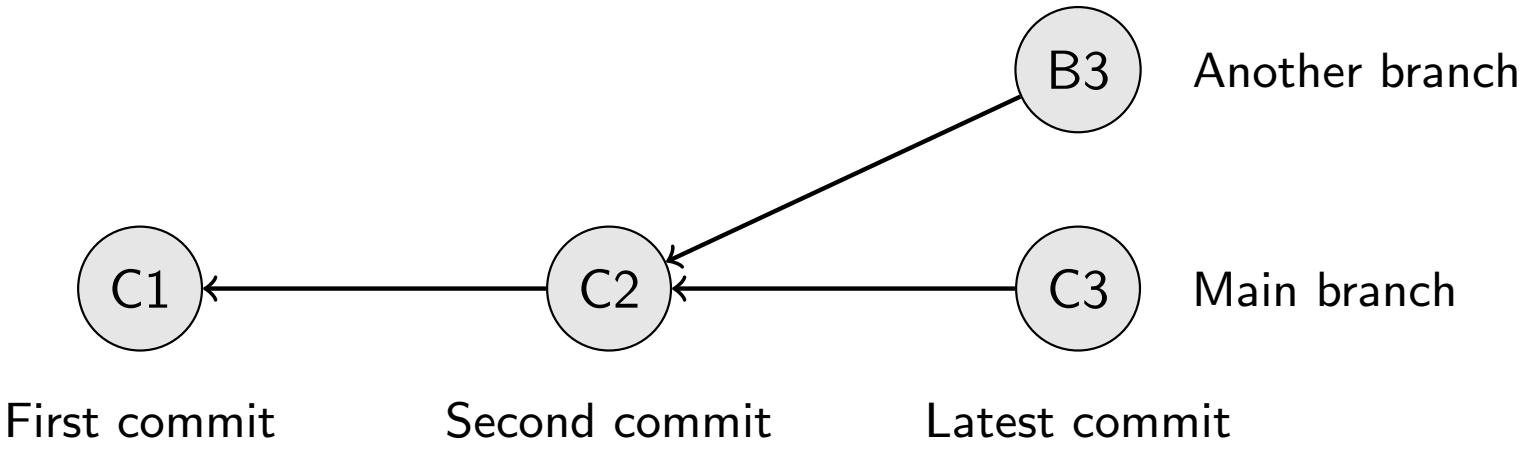
```
\begin{tabular}{ccc}
  \toprule
  Variable & Description & Value \\
  \midrule
  \( \alpha \) & Intercept & 0.5 \\
  \( \beta \)  & Slope     & 1.2 \\
  \bottomrule
\end{tabular}
```

Rendered Output

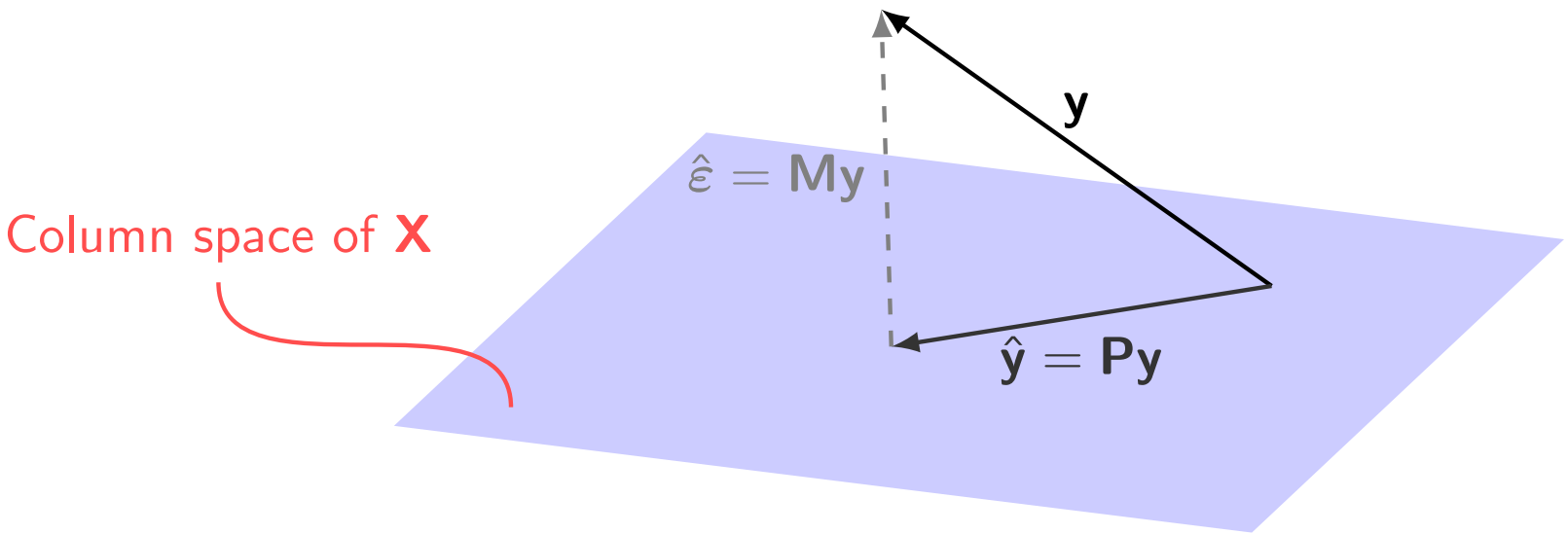
Variable	Description	Value
α	Intercept	0.5
β	Slope	1.2

Including Figures and Plots in LaTeX

Example 2D Figure



Example 3D Figure



Where to Write LaTeX?

- **Visual Studio Code:** A powerful code editor
  - LaTeX Workshop extension
  - Live preview
  - Git integration
  - Customizable environment
- **Overleaf:** An online LaTeX editor
  - No setup required
  - Real-time collaboration
  - Built-in templates
  - Access anywhere
- **Key Benefits:**
  - VSCode: More control and offline work
  - Overleaf: Easier to start and collaborate
  - Both support PDF compilation

Practical Tips for Beginners

- Start with simple documents and gradually add complexity
  - Begin with basic text and sections
  - Add equations and tables as needed
  - Learn advanced features gradually
- Use templates available online
  - Journal article templates
  - Presentation templates
  - Thesis templates
- Practice by typesetting your written homework to LaTeX.
- Practice typing math equations by: `https://texnique.xyz/`

Resources and Further Reading

- **Markdown Guide:** `https://www.markdownguide.org`
- **LaTeX Project:** `https://www.latex-project.org`
- **Overleaf:** `https://www.overleaf.com`
- Additional tutorials and documentation are available online

Summary

- Markdown offers a simple syntax for quick documentation to be rendered on the web.
- LaTeX is a powerful tool for producing professional, technical documents (especially PDF format).